UNITED STATES
NUCLEAR REGULATORY COMMISSION
WASHINGTON, D.C. 20555-0001

April 2, 2020

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Cranberry Headquarters
1000 Westinghouse Drive
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SUBJECT: NUCLEAR REGULATORY COMMISSION VENDOR INSPECTION REPORT OF WESTINGHOUSE – NEW STANTON, NO. 99901043/2020-201

Dear Mr. Harper:

From February 18 through February 21, 2020, the U.S. Nuclear Regulatory Commission (NRC) staff conducted an inspection at Westinghouse's (hereafter referred to as W-NS) facility in New Stanton, Pennsylvania. The purpose of this limited-scope inspection was to assess W-NS's compliance with the provisions of Title 10 of the Code of Federal Regulations (10 CFR) Part 21, "Reporting of Defects and Noncompliance," and selected portions of Appendix B, "Quality Assurance Program Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to 10 CFR Part 50, "Domestic Licensing of Production and Utilization Facilities."

This technically-focused inspection specifically evaluated W-NS's implementation of the quality activities associated with testing of safety-related components that comprise the Uninterruptible Power Supply (UPS) system for the AP1000 reactors currently under construction at Vogtle Units 3 and 4. This NRC inspection report does not constitute NRC endorsement of W-NS's overall quality assurance (QA) or 10 CFR Part 21 programs.

Based on the results of this inspection, the NRC inspection team found the implementation of your QA program met the requirements imposed on you by your customers or NRC licensees. No findings of significance were identified.

In accordance with 10 CFR 2.390, "Public Inspections, Exemptions, Requests for Withholding," of the NRC's "Rules of Practice," the NRC will make available electronically for public inspection a copy of this letter, its enclosure, and your response through the NRC's Public Document Room or from the NRC's Agencywide Documents Access and Management System, which is accessible at http://www.nrc.gov/reading-rm/adams.html.

If you have any questions concerning this matter, please contact Mr. Aaron Armstrong of my staff at (301)415-8396.

EXECUTIVE SUMMARY

Westinghouse – New Stanton
99901043/2020-201

The U.S. Nuclear Regulatory Commission (NRC) staff conducted a vendor inspection at the Westinghouse (hereafter referred to as W-NS) facility located in New Stanton, PA, to verify that it had implemented an adequate quality assurance (QA) program that complies with the requirements of Appendix B, "Quality Assurance Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to Title 10 of the Code of Federal Regulations (10 CFR) Part 50, "Domestic Licensing of Production and Utilization Facilities" and 10 CFR Part 21, "Reporting of Defects and Noncompliance."

Translation of Technical Requirements

The NRC inspection team reviewed W-NS's translation of requirements for the Environmental Qualification (EQ) testing of safety-related uninterruptable power supplies (UPS's), Class 1E Direct Current (IDS) and UPS System Battery Charger, Inverter, and Regulating Transformer in accordance with NRC regulations and the technical standards imposed on them by their customer for components being supplied to the AP1000 units under construction at the Vogtle site. The NRC inspection team verified that the technical requirements contained within the purchase orders and design specifications were properly translated into W-NS EQ Testing Procedures. No findings of significance were identified.

Adequacy of Testing Scope

The NRC inspection team reviewed the overall scope of the EQ testing program for the UPS. In particular, the NRC inspection team assessed whether the testing program scope was adequate to demonstrate EQ of the required safety-related functions. The UPS safety-related functions consist of 1) supplying alternating current (AC) load 2) provide the Protection and Safety Monitoring System (PMS) with two loss-of-AC input voltage signals and 3) isolation between non-Class 1E and Class 1E circuits. The EQ testing scope was developed in accordance with the requirements of 1974 Editions of the Institute of Electrical and Electronics Engineers (IEEE) standard No. 323, "IEEE Standard for Qualifying Class 1E Equipment for Nuclear Power Generating Stations"; NRC Regulatory Guide 1.89, Revision 1, "Environmental Qualification of Certain Electric Equipment Important to Safety for Nuclear Power Plants," and AP1000 Equipment Qualification Methodology. During the week of the inspection the NRC inspection team did not review the isolation safety-related function in accordance with IEEE 384-1981, "Standard Criteria for Independence of Class 1E Equipment and Circuits," because it was outside of the scope of the EQ test program during the week of the inspection. No findings of significance were identified.

Test Control

The NRC inspection team reviewed W-NS's policies and implementing procedures that govern the implementation of its testing activities to determine compliance with the regulatory requirements in Criterion XI, "Test Control," of Appendix B to 10 CFR Part 50. The NRC inspection team reviewed pre-test site acceptance testing, test deviations and test non-conformances and the physical test setup for conducting EQ tests defined in W-NS EQ testing procedures and work instruction. The objective of the W-NS EQ testing profile consisted of three phases; EQ mild environments (Phase 1), abnormal temperature and humidity transients

(Phase 2), and supplemental environmental conditions at minimum temperature (Phase 3). During the inspection, W-NS only tested Cycle 1 of the Phase 2 for the EQ profile because W-NS's environmental chamber was not able to maintain temperature and humidity conditions in accordance with IEEE 650-1990, "IEEE Standard for Qualification of Class 1E Static Battery Chargers and Inverters for Nuclear Power Generating Stations," required for Phase 1 or maintain the elevated relative humidity levels of Phase 2, Cycle 2. W-NS will complete the remainder of the EQ testing at Clark Laboratory. No findings of significance were identified.

Control of Measuring and Test Equipment

The NRC inspection team reviewed W-NS's policies and implementing procedures that govern the implementation of its measuring and test equipment (M&TE) program to determine compliance with the requirements of Criterion XII, "Control of Measuring and Test Equipment," of Appendix B to 10 CFR Part 50. The NRC inspection team observed that the M&TE had the appropriate calibration stickers and current calibration dates, including the calibration due date. The NRC inspection team verified calibration of M&TE in W-NS's facility and reviewed the procurement methods used for the calibration and testing services for equipment used for the AP1000 battery charger, inverter and rectifying transformer. No findings of significance were identified.

REPORT DETAILS

1. Design Control

a. Inspection Scope

The NRC inspection team reviewed Westinghouse's (hereafter referred to as W-NS) policies and implementing procedures that govern the design control program to determine compliance with the requirements of Criterion III, "Design Control" of Appendix B to 10 CFR Part 50. The NRC inspection team also discussed the design control program with Westinghouse's management and technical staff. The attachment to this inspection report lists the documents reviewed by the NRC inspection team.

Translation of Technical Requirements

The NRC inspection team reviewed Westinghouse's translation of the testing requirements for EQ testing of the safety-related uninterruptable power supplies (UPSs) being supplied to the AP1000 units under construction at the Vogtle site. The UPS Class 1E Direct Current (IDS) provides reliable power for the safety-related equipment required for the plant instrumentation, control, monitoring and other vital functions needed for shutdown of the plant. In the event of a total loss of off-site and on-site alternating current (AC) power sources, the direct current (DC) batteries constitute the sources of electrical power for operation of the required DC and AC instrument UPS loads. The NRC inspection team also reviewed associated purchase orders and purchase order revisions for the EQ testing. The NRC inspection team reviewed the W-NS's developed test procedures to ensure the UPS AP1000 environmental profile and equipment qualification methodology requirements were appropriately translated from the design specifications.

The NRC inspection team reviewed design specification APP-DC01-Z0-001 which provided the technical requirements for the design of the Class 1E battery chargers that will be used in the IDS for the Westinghouse AP1000 plant. The NRC inspection team reviewed the design specification in APP-DT01-Z0-010 that provides the technical requirements for the Class 1E regulating transformers for the Westinghouse AP1000 plant. The regulating transformers provide a regulated backup source of AC power to the Class 1E Instrumentation and Control (I&C) Bus. The regulating transformers are powered from the diesel generator backed Motor Control Centers. The regulating transformers provide the required isolation between the non-1E and the Class 1E electrical systems. In addition, the NRC inspection team reviewed design specification APP-DU01-Z0-001 which provides the technical requirements for the design of the Class 1E inverters, static transfer and manual by-pass switches. The inverters, static transfer switches, and manual by-pass switches are provided to power instrument and control power buses. The static transfer switch transfers the loads to the backup source upon inverter failure. Also, the inverters, static transfer switches, and manual by-pass switches are classified as Class 1E and AP1000 Class C.

The NRC inspection team reviewed test plan EQ-TP-488-APP developed for the pre-seismic environmental qualification (EQ) test for the IDS battery charger, inverter and regulating transformer for the AP1000 plant. The NRC inspection team verified that the test plan met the requirements contained within Regulatory Guide (RG) 1.89, Revision 1, "Environmental Qualification of Certain Electric Equipment Important to Safety for Nuclear Power Plants," and the associated equipment design specifications. RG 1.89 endorses IEEE 323-1974 and detailed the testing requirements for the EQ testing. The NRC inspection team also

interacted with the product engineering group to review the implementation of W-NS's receipt inspection requirements for the equipment under test (EUT).

Adequacy of Testing Scope

The NRC inspection team reviewed the scope of the EQ testing program. In particular, the NRC inspection team assessed whether the testing program scope was adequate to demonstrate the EQ of safety-related functions under the Phase 2, Cycle 1 EQ test profile. The NRC inspection team reviewed the EQ test program in accordance with IEEE 323-1974 and AP1000 testing requirements. The UPS EQ test consisted of the ability to demonstrate the equipment performance under abnormal environmental conditions of temperature and humidity that may occur during plant operation for a required duration of time.

The NRC inspection team reviewed the UPS functional operation test in response to voltage change, frequency and loading. Specifically, the functional tests were performed at ambient conditions and included inverter to by-pass transfer, by-pass to inverter transfer, inverter full load rejection, inverter full load step, inverter load power factor test, charger undervoltage (UV) relay test, charger blocking diode test, and charger current limit test. In addition, the NRC inspection team reviewed the EQ temperature limits, general room group methodology, and GOTHIC (Generation of Thermal-Hydraulic Information for Containment) modeling heat structures for compartments in the AP1000 auxiliary building where the UPS configuration will be installed in accordance with W-NS's walk-in environmental chamber specifications used for the EQ test program. Furthermore, the NRC inspection team reviewed W-NS's and a contracted test lab's inter-works request scope of work, in support of the remaining required UPS EQ test program.

The attachment to this inspection report lists the documents reviewed by the NRC inspection team.

b. Observations and Findings

No findings of significance were identified.

c. Conclusions

The NRC inspection team verified technical requirements contained within the purchase specification were being properly translated into EQ testing procedures. The NRC inspection team also determined that the overall scope of the EQ testing program was adequate to demonstrate environmental qualification of the required safety-related functions. No findings of significance were identified.

2. Test Control

a. Inspection Scope

The NRC inspection team reviewed W-NS's policies and implementing procedures that govern its testing activities to determine compliance with the regulatory requirements in Criterion XI, "Test Control," of Appendix B to 10 CFR Part 50. The NRC inspection team also discussed the test control program with W-NS management and technical staff.

Pre-test Site Acceptance Testing (SAT)

The NRC inspection team reviewed a contractor's SAT report to confirm the system was received and installed adequately. The SAT testing included a general condition check, component tests, alarm tests, and functional testing. All testing was adequately documented and confirmed proper installation and functioning of the system. During SAT, one issue was identified by W-NS with IDS battery charger high DC voltage warning and high DC voltage shutdown that required system setpoints to be revised. A design change was adequately implemented in accordance with the requirements for an Engineering and Design Coordination Report (E&DCR).

EQ Test Setup and On-Going Test Review

The NRC inspection team reviewed the physical test setup for conducting EQ tests defined in W-NS's EQ test procedures and work instructions. The W-NS procedures define the process and responsibilities for conducting EQ testing, general quality practices including, but not limited to, calibration and software testing, test area access, byproduct control, development of test procedures, test logs, reporting of nonconforming conditions, and final test report generation. In addition, the NRC inspection team performed a walk-down of the test configuration, observed EUT and discussed the ongoing test program with W-NS's testing staff.

The NRC inspection team reviewed the current environmental test log that included results of initial inspection, test specimen modifications (including field changes), and setpoint modifications. The test log also provided test configuration diagrams, and all monitoring equipment and connections to the EUT were identified. The NRC inspection team confirmed that monitoring channels were established for power, signal, and environmental data. The NRC inspection team confirmed the test setup and environmental conditions met the test configuration as defined in the test logs.

Review of Test Deviations and Non-Conformances

The NRC inspection team also reviewed the current sample of test deviations and associated nonconformances (NONs) from the ongoing Phase 2 testing and confirmed the vendor was documenting and evaluating any test anomaly issues identified with the implementation of the test procedure.

The NRC inspection team reviewed completed test logs sheets for six data cycles and confirmed the results were generally consistent with the expected min/max values for the parameters of interest. In instances of the recorded min/max values not meeting test specifications, an NON was created, identified on the test log, and documented in the nonconformance reporting system. The NRC inspection team reviewed a sample of NONs generated from the Phase 2, Cycle 1 tests and confirmed the descriptions, comments on dispositions, and resolutions proposed or implemented. In several instances, changes required to the test procedures were captured in deviation reports that were documented on the test log sheets. The NRC inspection team reviewed the deviation reports and confirmed the descriptions, comments, and dispositions were consistent with those changes described in the deviation and/or the test logs.

The attachment to this inspection report lists the documents reviewed by the NRC inspection team.

b. Observations and Findings

No findings of significance were identified.

c. Conclusions

The NRC inspection team concluded that W-NS was implementing its test control program in accordance with the regulatory requirements of Criterion XI of Appendix B to 10 CFR Part 50. The NRC inspection team verified test control and work instructions requirements were being properly controlled for EQ testing. No findings of significance were identified.

3. Control of Measuring and Test Equipment

a. Inspection Scope

The NRC inspection team reviewed W-NS's policies and implementing procedures that govern the M&TE program to determine compliance with the requirements of Criterion XII, "Control of Measuring and Test Equipment," of Appendix B to 10 CFR Part 50. The NRC inspection team also discussed the M&TE program with W-NS management and technical staff.

The NRC inspection team reviewed the calibration records for the SOLTEC data acquisition modules, thermocouples, Fluke power analyzers, and Rotonic data loggers. The NRC inspection team also reviewed W-NS's purchase orders for a contracted test laboratory. The NRC inspection team observed that the M&TE had the appropriate calibration stickers and current calibration dates, including the calibration due date. In addition, the NRC inspection team reviewed purchase orders for M&TE calibration services and testing services. The NRC inspection team performed a walk-down of W-NS's testing laboratory to observe that M&TE were labeled, handled, and stored in a manner that indicated the calibration status of the instrument and ensured its traceability to calibration test data.

The attachment to this inspection report lists the documents reviewed by the NRC inspection team

b. Observations and Findings

No findings of significance were identified.

c. Conclusion

The NRC inspection team concluded that W-NS is implementing its M&TE program in accordance with the regulatory requirements of Criterion XII of Appendix B to 10 CFR Part 50. No findings of significance were identified.

4. Entrance and Exit Meetings

On February 18, 2020, the NRC inspection team discussed the scope of the inspection with Mr. Brian Schleger and other members of W-NS's management and technical staff. On

February 21, 2019, the NRC inspection team presented the inspection results and observations during an exit meeting with Mr. Brian Schleger, and other members of W-NS's management and technical staff. The attachment to this report lists the attendees of the entrance and exit meetings, as well as those individuals whom the NRC inspection team interviewed.